CYS334 Network Security

Practice Lab#1: Network Topology Implementation for ABC Company

1. Lab Title

Network Topology Implementation for ABC Company (Two-Branch Network)

2. Lab Objectives

By the end of this lab, students will be able to:

1. Design and implement a network topology connecting two branches via routers.
2. Configure VLANs and trunk links on switches.
3. Assign IP addressing using a suitable IP addressing scheme.
4. Configure inter-VLAN routing using routers.
5. Activate and verify **DHCP service** on routers to dynamically assign IP addresses to hosts in each VLAN.
6. Verify end-to-end connectivity between branches.

3. Scenario / Background

ABC Company has **two branches**, separated by approximately **200 km**. Each branch requires:

- 2 switches,
- 3 VLANs (Admin, Staff, Operation),
- Inter-VLAN communication via router,
- Dynamic IP assignment via DHCP,
- Inter-branch communication using routers.

The IT department is tasked with building and configuring this network to ensure full functionality between branches.

4. Network Topology Overview

Devices Required

Branch	Routers	Switches	PCs (per VLAN)
Branch 1	1 × Router1	2 × Switches (S1, S2)	3 PCs per VLAN
Branch 2	1 × Router2	2 × Switches (S3, S4)	3 PCs per VLAN

Connection

- Routers (R1 ↔ R2) connected via **Serial link** (WAN).
- Switches connected with **Trunk links**.
- PCs connected to access ports of their corresponding VLAN.

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5. IP Addressing Scheme

Branch	Network	VLAN	Subnet	Default Gateway
Branch 1	10.10.10.0/24	VLAN 10 (Admin)	10.10.10.0/26	10.10.10.1
		VLAN 20 (Staff)	10.10.10.64/26	10.10.10.65
		VLAN 30 (Operation)	10.10.10.128/26	10.10.10.129
Branch 2	20.20.20.0/24	VLAN 10 (Admin)	20.20.20.0/26	20.20.20.1
		VLAN 20 (Staff)	20.20.20.64/26	20.20.20.65
		VLAN 30 (Operation)	20.20.20.128/26	20.20.20.129

WAN Link

Interface	IP Address
R1 Serial0/0	192.168.100.1/30
R2 Serial0/0	192.168.100.2/30

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6. Lab Tasks

Task 1: Topology Construction

1. Place **2 routers (R1, R2)**, **4 switches**, and **18 PCs (9 per branch)**.
2. Connect devices as follows:
    - Router ↔ Switch1 (GigabitEthernet interface)
    - Switch1 ↔ Switch2 (Trunk link)
    - Switches ↔ PCs (Access links)

Task 2: VLAN Configuration

Write the required VLAN configuration on each branch's switches:

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Task 3: Router-on-a-Stick Configuration (Inter-VLAN Routing)

Write the required configuration on **R1 (Branch 1)**:

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On **R2 (Branch 2)**:

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Task 5: Configure Routing Between Branches

Write the required Routing configuration ,Use static routes (for simplicity):

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Task 6: Configure DHCP on Routers

Write the required configuration to activate DHCP service on Branch 1 Router (R1):

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Branch 2 Router (R2):

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7. Verification

- Use show ip interface brief to confirm IP assignments.

- Check DHCP bindings using show ip dhcp binding.

- Ping between PCs in different VLANs within the same branch.

- Ping from a PC in Branch 1 to a PC in Branch 2.

- Test DHCP assignment on multiple PCs.

8. Deliverables / Student Submission

Students must provide:

1. Completed Packet Tracer (.pkt) file.
2. Screenshot evidence of successful pings between:
    - VLANs within a branch.
    - Branch 1 ↔ Branch 2 PCs.
3. A brief report explaining: (single paragraph of 5 lines at most)
    - VLAN configuration summary.
    - Routing configuration.
    - DHCP function verification.
    - Main security notice of this topology

- Packet tracer Download

  CiscoPacketTracer_900_win_64bit.exe